University of Mumbai

Bomb Detection System using IoT

Submitted in partial fulfillment of requirements

For the degree of

Bachelor of Technology

by

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Batch 2018-2022

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Certificate

This is to certify that the dissertation report entitled **Bomb Detection System using IoT** is abonafide record of the dissertation work done by Ayush Parmar, Jas Prajapati, Heeral Gawade, and Yash Dave in the year 2018-2022 under the guidance of Dr. Vandana Satam of Department of Electronics and Telecommunication Engineering in partial fulfillment of the requirement for the Bachelor of Technology degree in Electronics and Telecommunication Engineering of University of Mumbai.

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Abstract

Nearly every day, a significant number of news stories about wounded trained people or military personnel who die while trying to defuse explosives appear in the news everywhere globally. Thus, a system is built to identify a threat that is positioned within a 1m range as well as to protect the bomb disposal team from danger. The user can control the developed system manually using a Private Computer (PC) or a mobile device. When the sensor detects a metal, the buzzer sounds an alarm. The metal is analyzed using a wireless camera to determine whether or not it is a bomb. If indeed the detected metal turns out to be a bomb, the user uses a PC to operate the System and set off the explosives. The developed System has an Arduino NANO board, DC motors, system arms, a wireless camera, and a buzzer. The arrangement is first simulated with Proteus application, and then the full hardware configuration is managed via a personal computer.

Our project aims to create a wireless system with a metal detector for bomb surveillance and detection. Using 500 RPM geared motors, it can be moved forward as well as backward. In addition, this System can make sharp turns to the left and right. The Arduino NANO serves as the project's controller. This System is equipped with a high-capacity metal detector circuit. If a bomb is detected when the System moves on a surface, the system emits a beep sound. This project will be extremely beneficial in the detection and surveillance of mines.

***Keywords*:** Bombs, a buzzer, Arduino NANO board, trained crew, Personal Computer, Proteus

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Nomenclature

L	Inductance in Henries
μo	Permeability is $4\pi*10-7$ for Air
N	Number of turns
A	Inner Core Area ($\pi r2$) in m2
l	Length of the Coil in meters

Chapter 1

Introduction

This chapter presents a brief introduction about the background of the project we choose and what tempted us to choose this project, the scope, a brief description, and the organization of the project.

1.1 Background

A System is a mechatronic device that can be productive or independent. A gadget having autonomy can function "on its own" without the intervention of a human.

Robotics is indeed the apex of technological growth at the present. Robotics is a hybrid science that combines mechanical engineering, material science, sensor fabrication, production processes, and complex algorithms. For some, robotics romanticism inspires an almost magical curiosity about the world, leading to the creation of incredible machines. In the field of robotics, you're in for a once-in-a-lifetime adventure.

Humans have been the driving force behind the development of System systems for many years. The Systemic research that took place thus far, had to face many problems and restrictions. Several attempts have been made to develop systems to be utilized as a replacement for humans to perform duties that are a little more critical and harmful to human life, such as bomb disposal, machine cutting, and so on.

Our main goal is to develop a system that can be used for bomb dispersion and disposal and can be controlled remotely by a bomb disposal expert, with the device transported by the System's robotic arm and safely disposed off away from human populations. We are going to build a system that can both command and control. The user's commands are received in the form of control signals, and the necessary action is carried out. The aim behind this is to give a bomb disposal team a line of protection against the life-threatening risk they confront in the case of an explosion.

1.2 Motivation

Robotics is the science or study of technology that is largely concerned with the design, manufacture, theory, and application of Systems. Robotics generates the magical final product, whereas other professions provide the math, procedures, and components.

The System's main motivation is to assist the bomb disposal squad in providing protection and security from the hazards they face regularly. The System's functioning is controlled via a wireless module, which allows for a greater range of operation. Also to construct a simplistic bomb disposal System that can accomplish simple tasks such as cutting wires, turning on switches, lifting light things, and so on before approaching it fordisposal.

1.3 Scope of the Project

A mobile System reduces or eliminates the time a bomb technician spends on target. A System removes the element of danger from potentially deadly situations, allowing the bomb specialist to concentrate on what to do with an explosive device rather than the immediate threat to life and limb. As a result, it will undoubtedly aid in the survival of those involved in bomb detection and dissemination.

1.4 Brief description of the Project

The System's principal purpose is to assist the bomb detection and disposal team by providing protection and security from the threats they face on a regular basis. A wireless module controls the System's operation, allowing for a larger range of operations like constructing a basic bomb disposal system that can handle simple tasks such as cutting wires, turning on switches, lifting light things, and so on. A mobile system minimizes or eliminates the time spent on target by a bomb technician. A System takes the risk out of potentially deadly scenarios and lets the bomb technician focus on what to do to an explosive device rather than on the immediate danger to life and limb. Even if a System cannot reach an item for disruption, it can still be used to relay information to aid in tool and procedure selection to move downrange. Following are some of the features:-

- An Android phone will be used to control the system. This user input will be sent serially to the System, where it will be received, identified, and further instructed by the System module.
- The user will be the source of input in this project.
- The output of the system is then processed into a signal that can instruct the appropriate module.
- This module will be a motor for the wheels of the System.
- Because of its small weight, it can operate in places where no human can, such as across land mines.
- Easy to control.

1.5 Organization of Report

- The 1st chapter of the report gives a brief introduction, motivation, scope, and explanation of the project.
- The 2nd chapter of the report gives the bomb detection-related information used for this project.
- The 3rd chapter of the report gives the problem argument, a block diagram, and an objective for the study.
- The 4th chapter of the report gives the implementation process and analysis in brief
- The 5th chapter gives the conclusion and future works part of the project.
- The 6th chapter gives the references used

Chapter 2

Literature Survey

This chapter's main focus was on the research process to find all the papers in which "IoT", "Bomb Detection" and "Metal Detection" have been mentioned. The search process has been carried out on IEEEXplore and different data base. Thus, we have presented overall research done in Bomb Detection using IoT till date.

In today's world, bomb detection, disposal, and diffusing have become a critical and dangerous way of human life, therefore a proposal based on a wireless bomb disposal System has been developed to solve the problem. The bomb will be destroyed by a System that will be controlled via a wireless control module. Currently, a variety of tools are widely used to detect explosives. From dogs and honeybees to mass spectrometry, gas chromatography, sound waves, and specially developed X-ray equipment, there's something for everyone.

Bluetooth technology is the primary method of serial communication with the System that we have employed here. Given the range between two devices, Bluetooth technologycan be utilized to share data between them. The System will be connected to the Bluetooth module HC-05, and commands will be delivered to the System via the android application. The four key components we will use in this project are:

2.1 Bluetooth Module HC-05

This Bluetooth SPP (Serial Port Protocol) module is designed for the development of a transparent wireless serial connection. It also offers a mode that allows you to switch between devices. The Bluetooth module has two devices: one is the master device, and the other is the slave device. The master is connected to one device, while the slave is connected to the other. The HC-05 Bluetooth module has six pins. The six pins are Key, 5V, GND, Tx, Rx, and Status. The master device must be linked to the slave for proper communication. When two devices are paired, the device will prompt you to enter your password. Either 0000 or 1234 will be the password. Once you've entered the password, both devices will be connected.

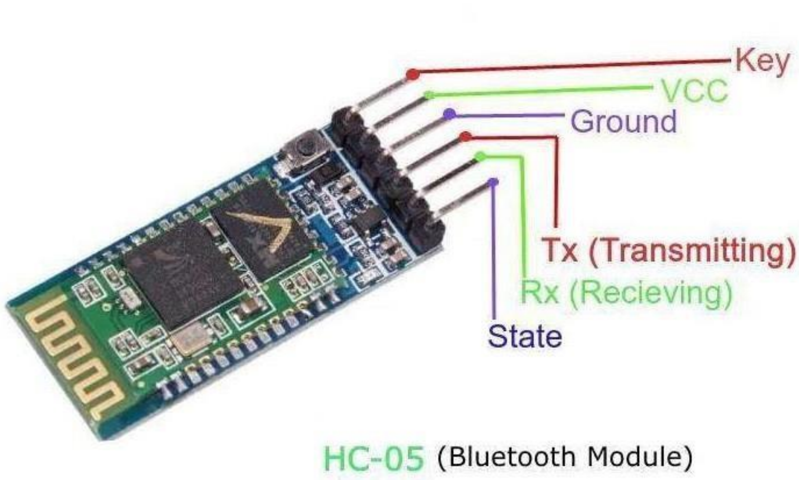


Fig. 2.1 HC-05 Bluetooth Module

2.1.1 HC-05 Specifications:

- Bluetooth protocol: Bluetooth Specification v2.0+EDR
- Dimension: 26.9mm x 13mm x 2.2 mm
- Emission power: ≤4dBm, Class 2
- Frequency: 2.4GHz ISM band
- Modulation: GFSK(Gaussian Frequency Shift Keying)
- Power supply: +3.3VDC 50mA
- Profiles: Bluetooth serial port
- Security: Authentication and encryption
- Sensitivity: ≤-84dBm at 0.1% BER
- Speed: Asynchronous: 2.1Mbps(Max) / 160 kbps, Synchronous: 1Mbps/1Mbps
- Working temperature: -20 ~ +75Centigrad

2.2 L293X Motor Driver IC

At voltages ranging from 4.5 V to 36 V, the L293X is designed to generate bidirectional drive currents of up to 600 mA. Both the devices 1,2 EN and the drivers 3 and 4 enabled by 3,4 EN can be used to drive inductive loads such as relays, solenoids, DC, and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications. When the input is high, the outputs are active and in phase with their inputs. Those drivers are disabled when the enable input is low, and their outputs are off and in the high-impedance condition. A full-H (or bridge) reversible drive with the necessary data inputs formed by each pair of drivers is suitable for solenoid or motor applications.

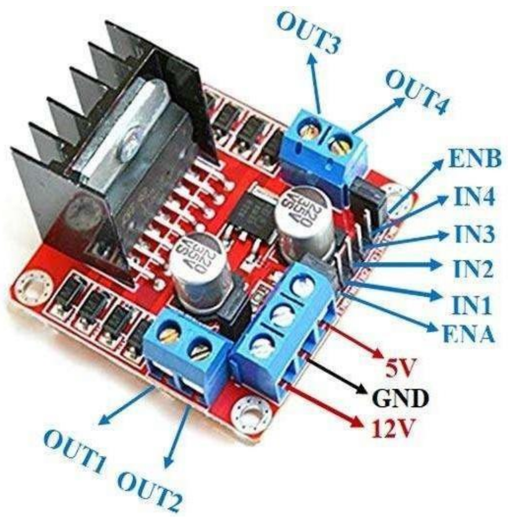


Fig.2.2 L298N Motor Driver Module

2.2.1 Features of the L298N Motor Driver Module:

- Operating source voltage up to 46 V.
- Total DC up to 4 A. 2A per channel.
- Saturation voltage is low.
- Protection from extreme temperatures.
- Logical input voltage up to 1.5 V.
- High noise immunity.

General Specifications of L298N:

Motor driver:	L298N
Motor channels:	2
Maximum operating voltage:	46 V
Peak output current per channel:	2 A
Minimum logic voltage:	4.5 V
Maximum logic voltage:	7 V
Package:	Multiwatt15

Table2.1. General Specifications of L298N Motor Driver Module

2.3 Arduino UNO

Arduino UNO contains everything needed to support the microcontroller along with 14 digital input/output pins; simply connect it to a computer with a USB cable or power it with an AC-to-DC adapter or battery to get started. It differs from all preceding boards because in that it does not use the FTDI USB-to-serial driver chip. It contains everything needed to support the microcontroller. We either need to connect it to a computer using a USB cable or power it with an AC-to-DC (7-12v) adapter. Arduino had used the Atmel Atmega AVR series of chips, specifically the ATmega8, ATmega168, ATmega328, ATmega1280, and ATmega 2560.

Arduino Uno	**Specifications**
Microcontroller	Atmega328P
Architecture	AVR
Operating Voltage	5 Volts
Flash Memory	32 KB of which 2 KB used by Bootloader
SRAM	2KB
Clock Speed	16 MHz
Analog I/O Pins	6
EEPROM	1 KB
DC Current per I/O Pins	40 milliAmps
Input Voltage	(6-20) Volts
Digital I/O Pins	8
PWM Output	6
Power Consumption	19 milliAmps
PCB Size	18 x 45 mm
Weight	7 gms

Table2.2. General Specifications of Arduino UNO

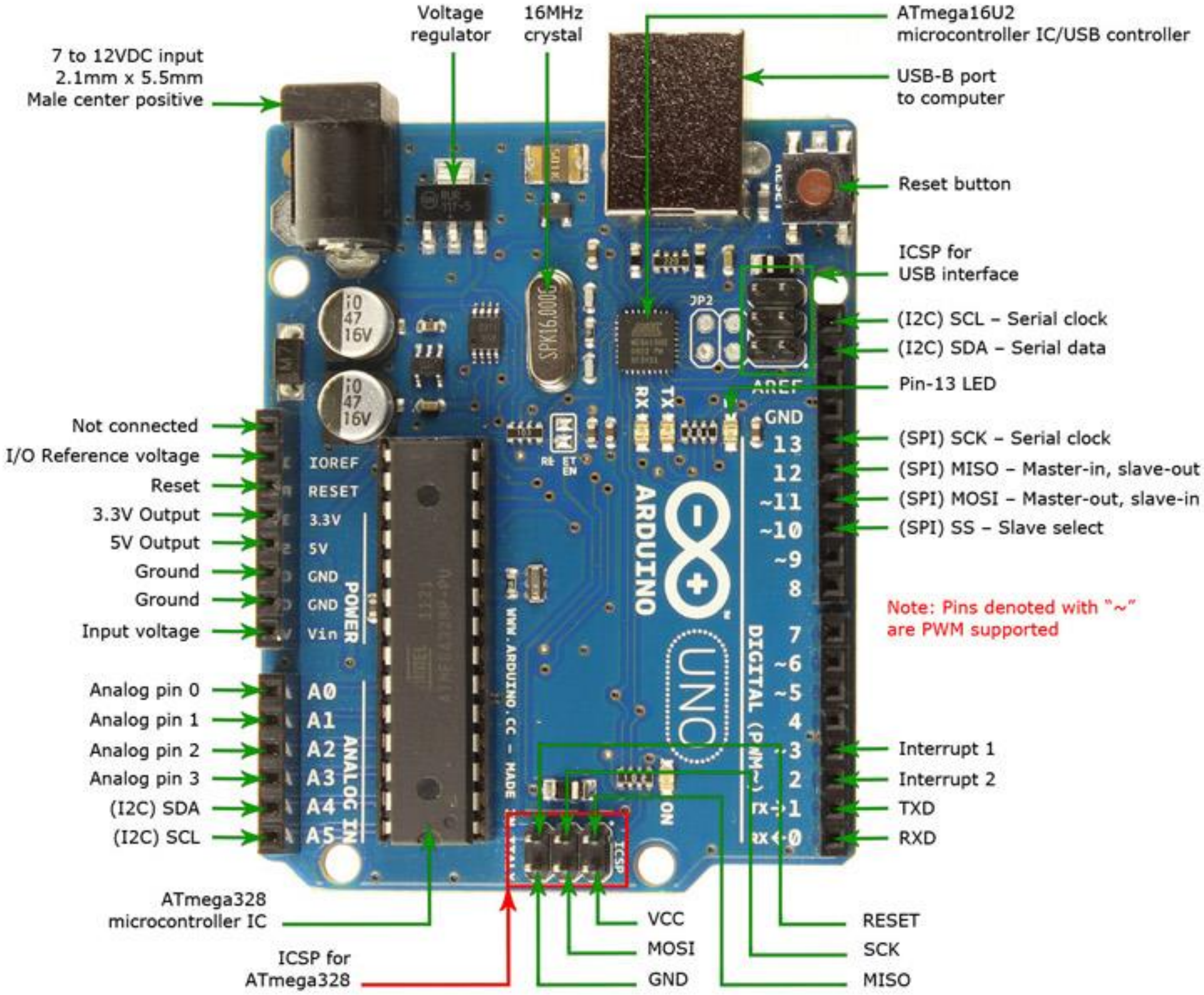


Fig.2.3 Arduino UNO Pin Description

2.4 DC Motor

Fig.2.4 RF-500TB DC Motor

A DC motor is an electric machine that converts electrical energy into mechanicalenergy. DC motors use direct current to convert electrical energy into mechanical rotation.

The movement of a rotor positioned within the output shaft is powered by magnetic fields induced by electrical currents in DC motors. The electrical input, as well as the motor's design, influence the output torque and speed.

Motor parameters

MODEL	VOLTAGE		NOLOAD		LOAD		STALL	
	OPERATING RANGE	Standard Voltage	SPEED (r/min)	CURRENT (A)	TORQUE (N．m)	OUTPUT (W)	TORQUE (N.m)	CURRENT （A）
RF-500TB-14415	3.0-12.0V	6.0V	3700	0.028	1.37	0.43	7.15	0.68

Table2.3. Parameters of RF-500TB-14415 DC Motor

2.5 Buzzer

An electric coil is wound on a plastic bobbin, which has a central sleeve into which a magnetic core is slide-able. The sleeve's one end is closed and extends beyond the coil. The coil and bobbin are encased in inverted cup-shaped housing with a central aperture through which the sleeve's closed-end projects.

To improve magnetic coupling between the housing and the core, the core projects into the closed end of the sleeve beyond the margin of the aperture in the housing. The open end of the housing is coupled to a magnetic support bracket, with a spring between the bracket and the bobbin generally forcing the core toward the sleeve's closed end.

Fig.2.5 Buzzer

2.6 Metal Detector

Metal detectors employ electromagnetic induction to detect metal. Metal detectors can assist you in locating metals buried deep underground. Demining (the detection of land mines), the detection of weapons such as knives and guns, especially at airports, geophysical prospecting, archaeology, and treasure hunting are just a few of the uses. Metal detectors are also utilized in the construction industry to detect foreign bodies in food, steel reinforcing bars in concrete, and pipelines and wires buried in blocks and floors.

In its most basic form, a metal detector consists of an oscillator that produces an alternating current that runs through a coil that produces an alternating magnetic field. When a piece of electrically conductive metal is placed near the coil, eddy currents are generated, causing the metal to produce an alternating magnetic field. If a different coil isused to measure the magnetic field, the change caused by the metallic object can be seen (serving as a magnetometer).

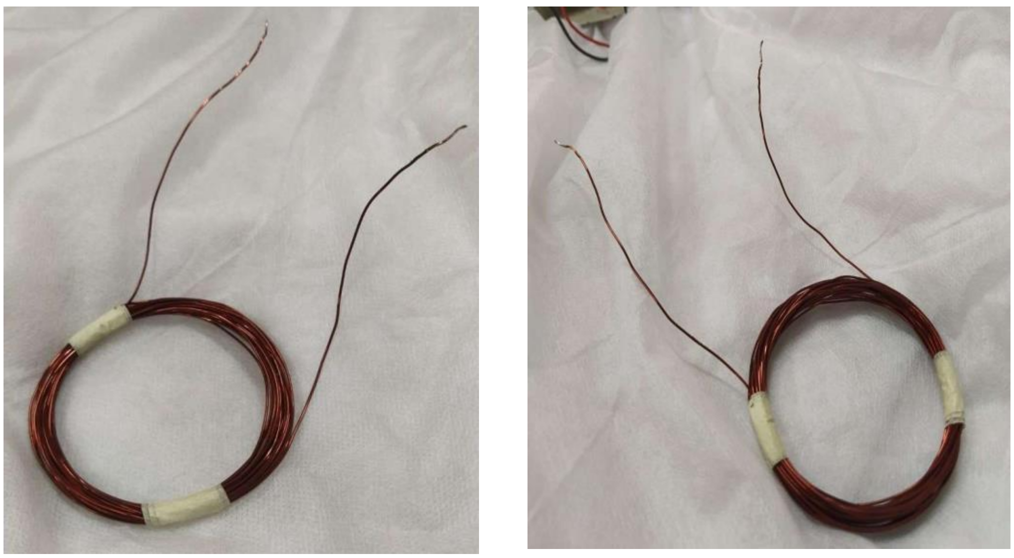


Fig.2.6 Electromagnetic Coil

2.7 Software

Arduino scripts are used to control the movement and the spinning of motors in this project. The Arduino software IDE is cross-platform and offers a straightforward programming environment.

2.8 Application Used

In this project, we have used the Bluetooth RC Controller application. This software is used to communicate with the Bot, which is controlled by a smartphone. The app lets you operate the car using virtual buttons or the accelerometer on your phone. When the phone is connected to the automobile, a flashing light appears, and arrows illuminate to indicate the car's driving direction.

Fig.2.7 Bluetooth RC Controller

2.9 3D Printed Case

To make our bot look cleaner and neater, we used a 3D printed cover case for our project. The dimensions of the case are 18cm x 10.5cm x 4cm.

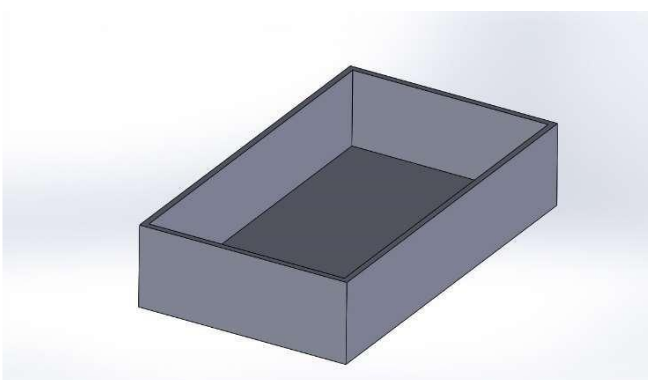


Fig.2.8 3D Printed case

Chapter 3

Project Design

This chapter gives the introduction of the project, the problem statement defined, the Flowchart process and the objectives to be achieved through this project is to minimize Human interaction with life-threatening operations.

3.1 Introduction

Our concept is based on the current state of civil wars, military instability, and terrorist scenarios around the world. Almost every day, a large number of trained personnel are hurt or killed when dealing with or attempting to disarm bombs. The limitless number of news items and films that surface every day on news channels and print media around the world can attest to all of this.

The robot's main purpose is to assist the bomb detection and disposal team in providing safety and security from the hazards they face regularly. The System's functioning is controlled via a wireless module, which allows for a greater range of operation.

A mobile System reduces or eliminates the time a bomb technician spends on target. A System removes the element of danger from potentially deadly situations, allowing the bomb specialist to concentrate on what to do with an explosive device rather than the immediate threat to life and limb. Even if a System is unable to disturb an item, it can still be utilized to convey information to aid in the selection of tools and procedures for going downrange.

3.2 Problem Statement

• To create an embedded system that will locate a metal body.

• Conduct research on the software and hardware that will be used.

• To write the code for the electrical circuit that has been assembled.

3.3 Flowchart

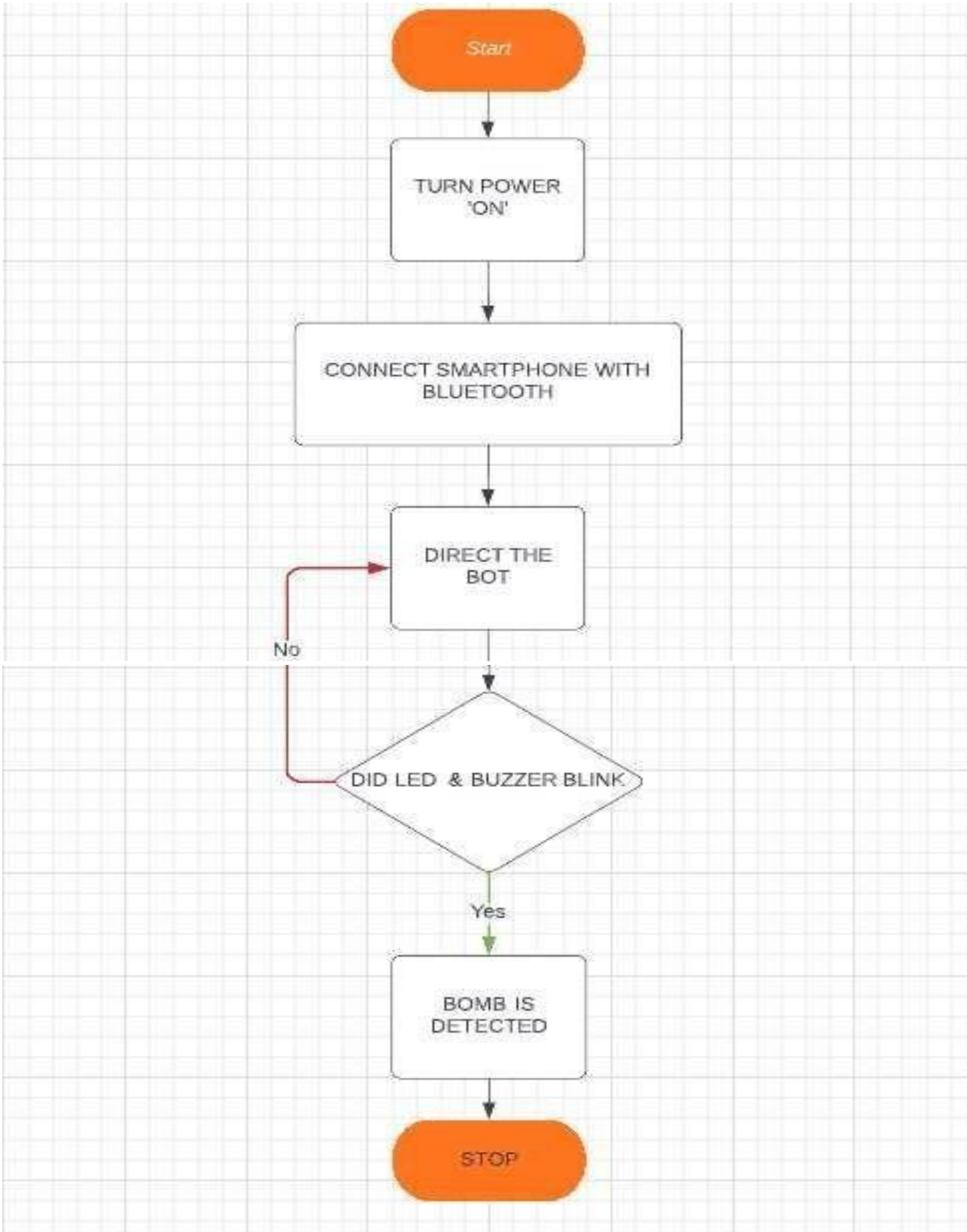


Fig.3.1 Flowchart of System

3.4. Objective

• In this project, we will build an embedded system that will be used to locate a metal body with the help of a System automobile controlled by the user's Android phone.

• The user will send data to move the automobile in any direction, and if any metal bodies are identified, a sound will be produced to alert the user.

3.5 Components Required

- Arduino NANO
- Bluetooth Module HC05
- 5V Power Supply
- 12V power supply for motor Driver
- Motor Driver (L298N)
- Digital Buzzer
- Android device
- Copper coil
- 1N4148 Diode
- 10nf capacitor
- Red Led
- 3X1Kohm Resistor
- USB cable for uploading the code

3.6 Circuit Diagram

3.6.1 Circuit Diagram for motor

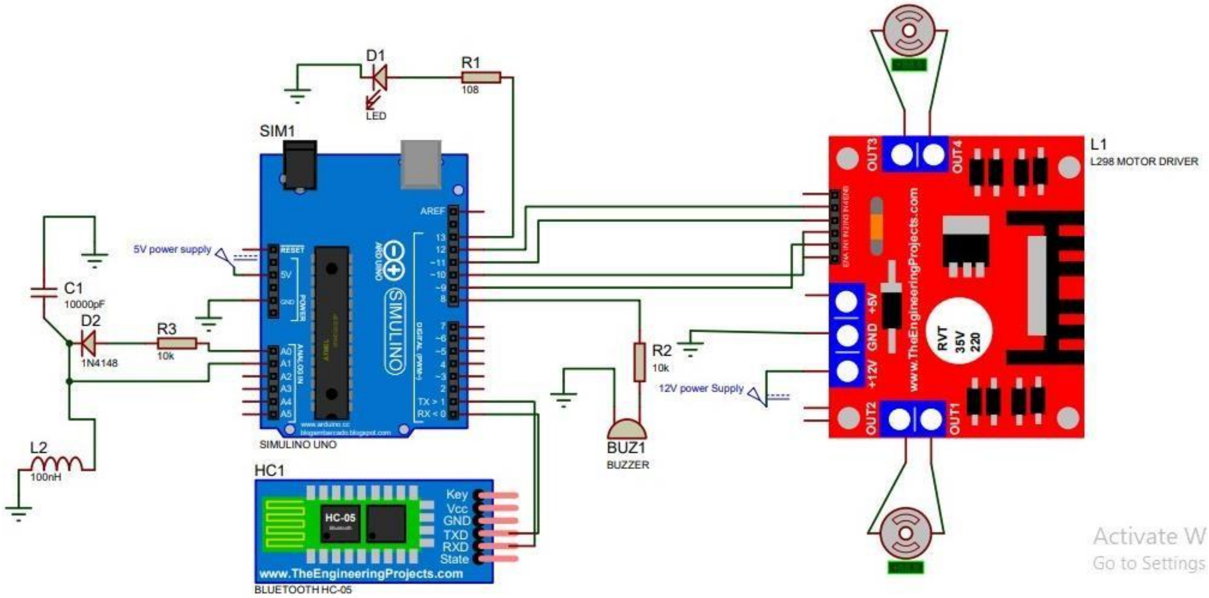


Fig.3.2 Motor Detection Circuit.

3.6.2 Working of Motor Circuit

- In this project, Arduino communicates with a variety of devices, including a motor driver (L298N) and a Bluetooth module (HC05).
- When the sensor detects a metal body within its range, it generates an analog value that is proportional to the distance between the sensor and the metal body.
- The sensor then communicates the analog value to the Arduino board through the A0 pin.
- After receiving an interrupt, Arduino sends data to motor driver pins 1,6,9,14, which are all set to LOW. As a result, both motors will come to a halt.
- An Arduino pin 9 HIGH is also created, which is connected to a buzzer. The buzzer will beep when it detects metal.

3.6.3 Circuit Diagram for Metal Detector

We created a metal detector circuit first to see if it is detecting adequately before creating our main circuit.

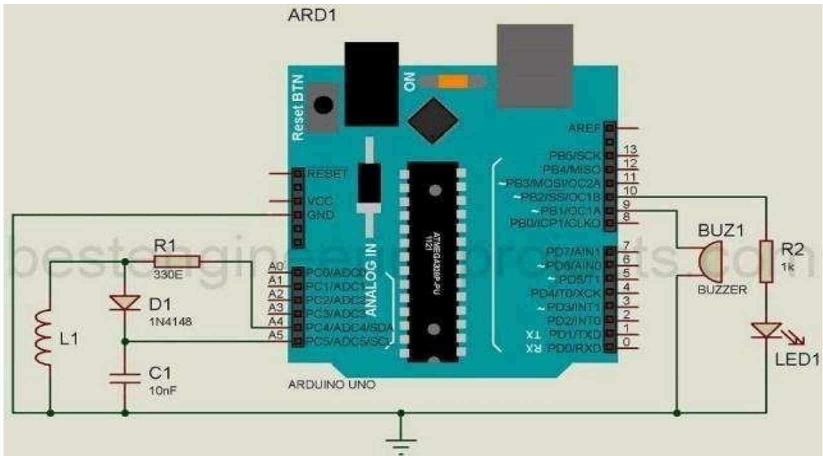


Fig.3.3 Simulated Circuit diagram of Metal Detector using Arduino

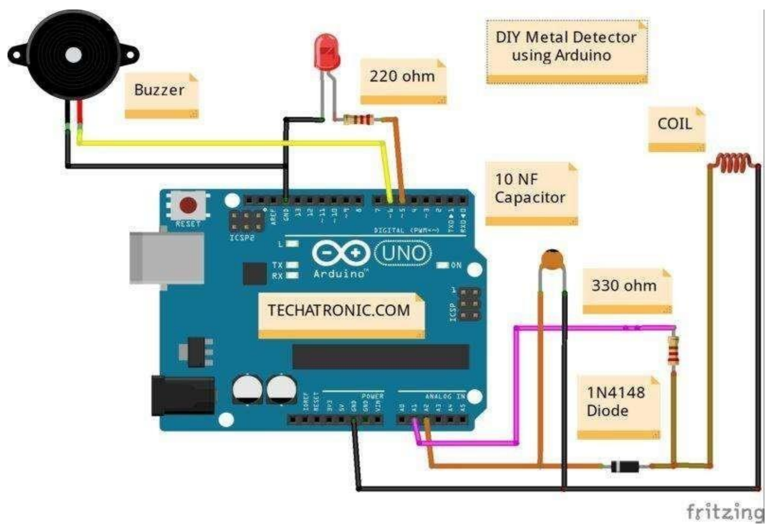


Fig.3.4 Metal Detector Circuit

Chapter 4

Implementation and experimentation

This chapter presents the process of implementation of our project. In this chapter, we discussed the steps of implementing, simulating, and assembling the circuit carefully as per our needs.

4.1. Steps for Implementation

- Research on the software applications and hardware components to be used.
- Proteus software was used to design the circuit.
- Using the arduino.exe software, try a few simple commands.
- Assembling the electrical circuit.
- Writing the Arduino code for the circuit.
- Pairing the hardware with software and testing the circuit.

4.2 Designed Circuit

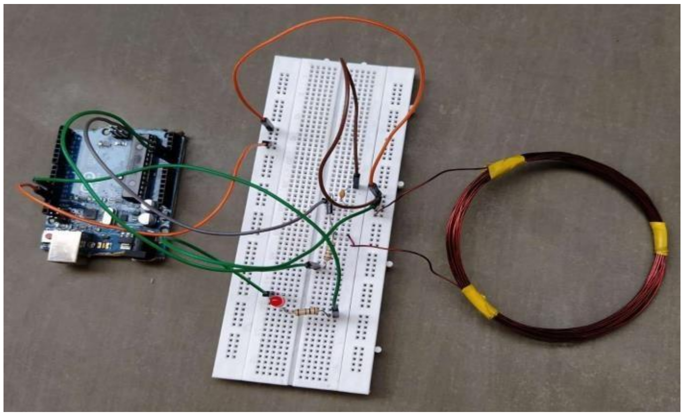


Fig.4.1 Designed circuit

The above metal detector circuit is made with an Arduino NANO board, a search coil, a buzzer, a 1N4148 diode, and a few other electronic components such as resistors, capacitors, and so on. The inner core of the coil determines the inductor's inductance. Inductor inductance is calculated using the following formula:

$$L = \frac{(\mu_0 \times N \times A)}{l}$$

Where,

L- Inductance in Henries

μo- Permeability, its 4π*10-7 for Air

N- Number of turns

A- Inner Core Area (πr2) in m2

l- Length of Coil in meters

4.2.1 Working of the circuit

We've utilized a parallel LC circuit in this example. Each reactive element (L and C) saves energy before releasing it to the other. The energy transfer between the twoelements will take place at a natural rate equal to the resonant frequency and will take theform of a sinusoidal wave. Because it lowers the voltage, the diode D1 is mostly employed as a resistor. R1 serves as a current limiting resistor, restricting the current to the Arduino pin. We used a buzzer and LED for the audio-visual effect. When metal is identified, both the buzzer and the LED are turned on.

The LR tank circuit is coupled to analog pins A4 and A5, which provide pulses and generate emf. The emf that is generated is stored in a capacitor. The voltage across the capacitor can be measured using the analog pin on the Arduino. Arduino reads the voltage, analyses it, and outputs it using the Analog to Digital Converter.

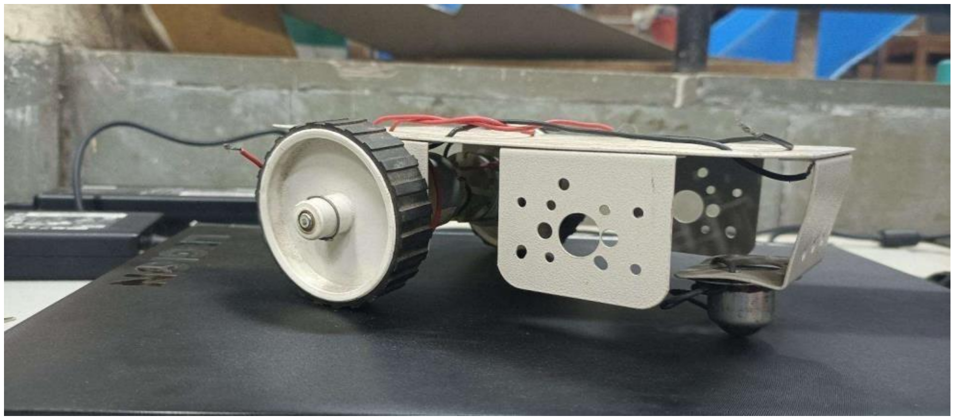
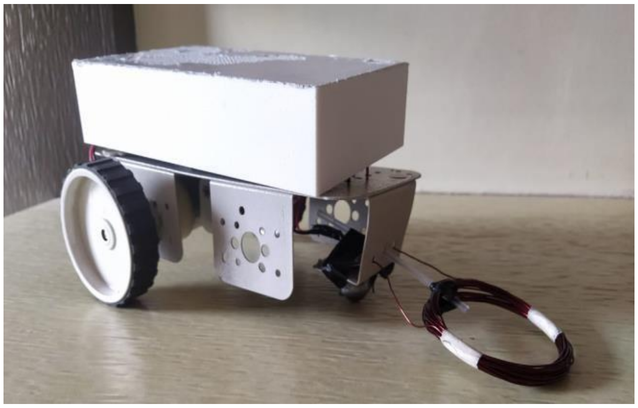


Fig.4.2 System Body

Fig.4.3 Final Circuit implemented on the bot



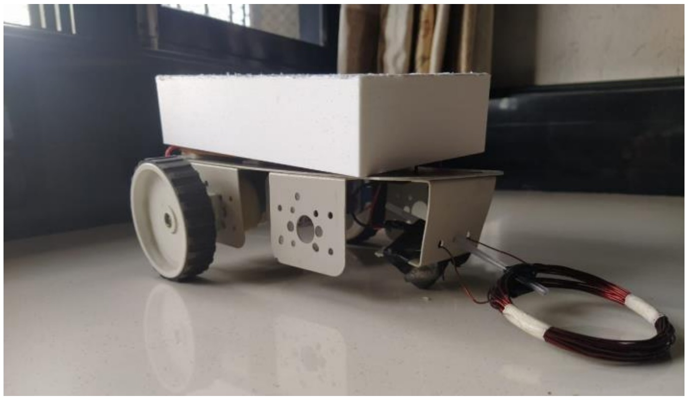


Fig 4.4 Final System

4.3 Procedure for Bluetooth RC Controller

i. Start the Bluetooth RC Controller on your Smartphone.

ii. Turn ON the Bluetooth on your Smartphone.

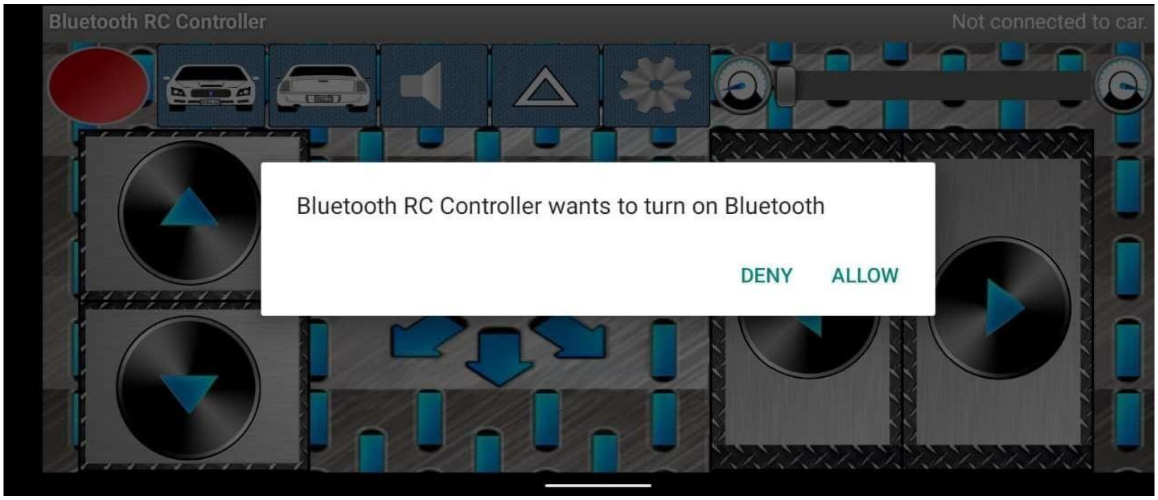


Fig.4.5 Bluetooth RC controller requests access

iii. Pair the device with Bluetooth Module HC-05

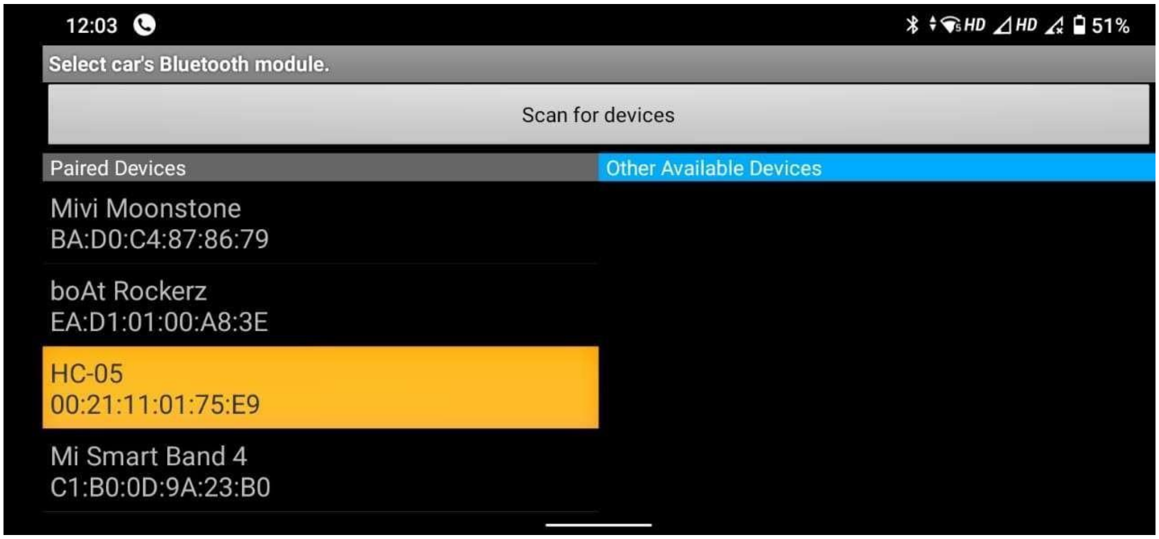


Fig.4.6 Pairing with the Bluetooth Module

iv. After successful connection, we use the displayed arrow keys for the movement of the Bot.

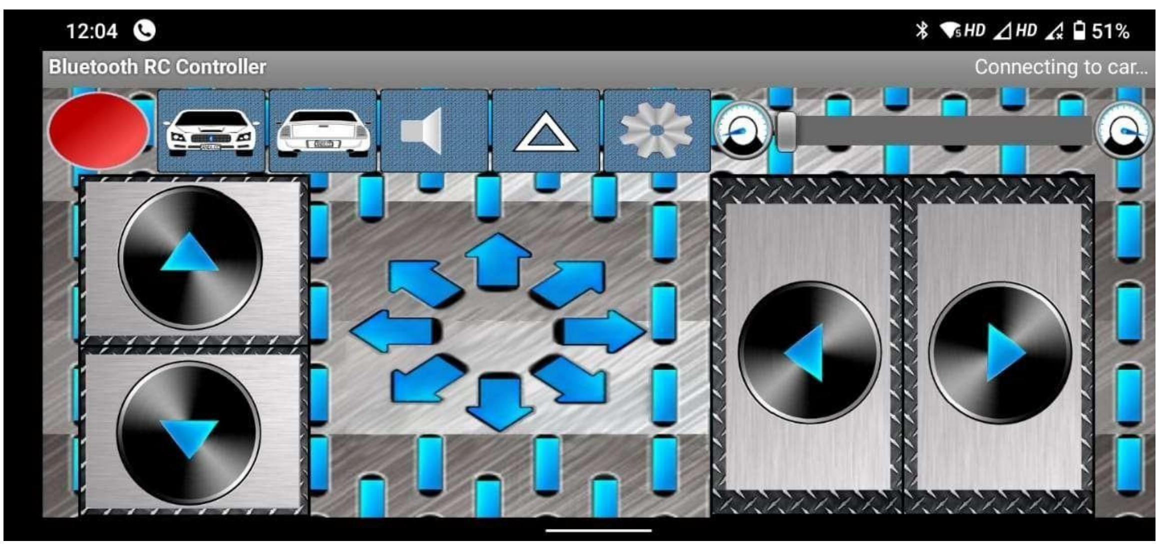


Fig.4.7 Pairing Successful

Here, after the connection is established between user and device, we use the arrow keys to move the system in the desired direction. The arrows glow in bright red color to signify the direction of movement.

4.2 Output Compilation

Fig.4.8 Compiled Output using the Code

The Code has been perfectly compiled as seen above.

Chapter 5

Results, Conclusions, and Scope for further work

This chapter discusses the results of the developed project, future work that can be done on this project to improve it, and the conclusion summarizes our learning from it.

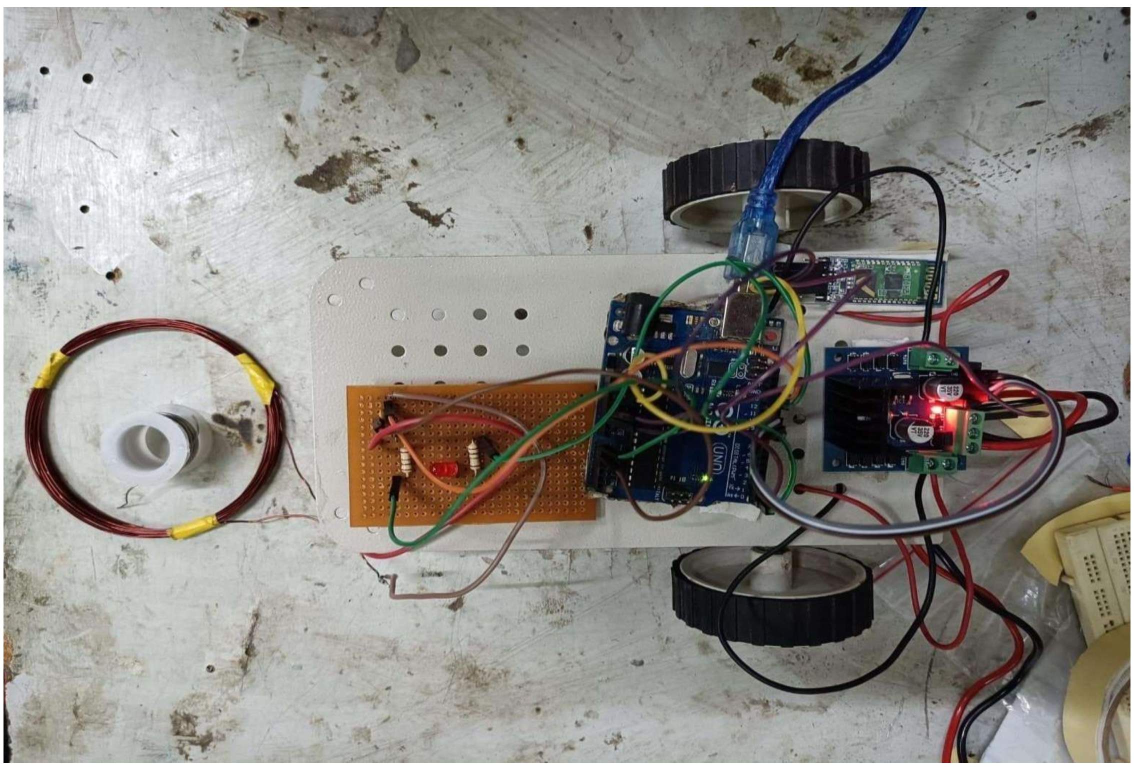


Fig.5.1 Circuit implemented on the bot

On our system, we have both the bot controller and the metal detector circuit shown in the diagram above.

5.1 Objects used to Obtain Results

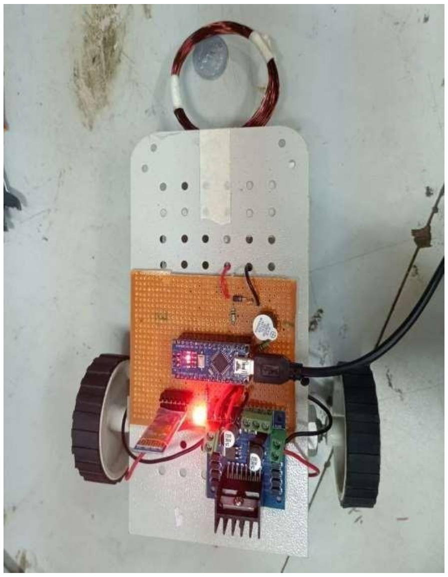
Fig.5.2 Objects used for detection

We have used a coin, wire stripper, a plastic scale, and a pen for detection purposes.

We have observed that during detection, the intensity of blinking of the LED depends on the volume and distance of metal from the coil.

Eg. The intensity increases (bright LED) when a wire stripper is detected and the intensity decreases (dim LED) for coin.

5.2 Results



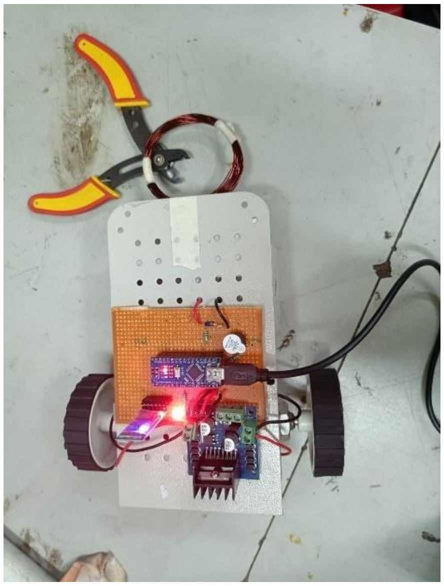
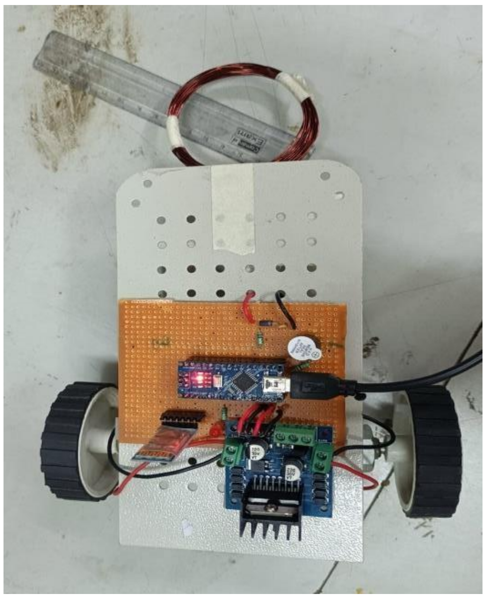
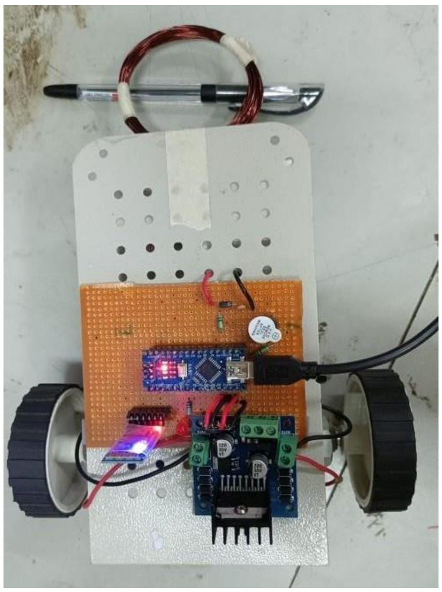


Fig.5.3 Wire stripper and Coin under the coil

In the above figures we can observe that when a wire stripper and a coin are placed under the coil, the LED blinks brightly.





In Fig.5.4 Pen and Scale under the coil

In the above figures, we can observe that when a pen and a scale are placed under the coil, there is no blinking of the LED.

From all the implementation and results we can summarize that,

Metal Content	***Frequency of LED Blinking***
LOW	LOW, DIM
MEDIUM	NORMAL
HIGH	HIGH, BRIGHT

Table5.1. Result summary

5.3 Conclusion

The wireless Bomb Detection System was created to meet the needs of the bomb disposal squad, the military, the police, and those who work with hazardous materials. It has a wide range of uses and can be employed in a variety of settings and scenarios. It can be employed by the bomb disposal team in some regions, and it can also be used to handle mines in others. As a result, the suggested approach gave us the experience in designing simple systems that benefit military applications. In our project, the user can control the system remotely using manual control. As a result, bomb disposal squads in the military and police could be replaced with specialized Systems.

5.4 Future scope

Additional features of this project could include the installation of a gas sensor, linking system arms for pick and place operations, and so on.

The system can't make decisions on its own right now because it lacks artificial intelligence. As a result, the task is entirely dependent on the end-user cases of the robotic control application. Therefore, artificial intelligence might be added to the System to make decision-making considerably faster and more dependable. The current system has a limited range of operations. As a result, this limitation may be improved in the future.

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Appendix A

Code

```
int m1a = 9;//for motor 1 forward
int m1b = 10;//for motor 1 reverse
int m2a = 11;for motor 2 forward
int m2b = 12;for motor 1 reverse

char val;
const byte npulse = 12; // chargig of cap using pulse measurement

const byte pin_pulse = A0; // use to send pulses to charge the capacitor (can
be a digital pin or analog pin)
const byte pin_cap  = A1; // use to measure the charged capacitor
const byte pin_LED = 13; // use to turn LED ON when metal is detected
const byte pin_buzzer=8;// use to ring the buzzer



void setup() {
   pinMode(m1a, OUTPUT); // set pin 9 as output Pin
  pinMode(m1b, OUTPUT); // set pin 10 as output Pin
  pinMode(m2a, OUTPUT); // set pin 11 as output Pin
  pinMode(m2b, OUTPUT);  // set pin 12 as output Pin

Serial.begin(9600);
  pinMode(pin_pulse, OUTPUT);//set pin as output Pin
  digitalWrite(pin_pulse, LOW);//set pin as LOW
  pinMode(pin_cap, INPUT);//set pin as INPUT Pin
  pinMode(pin_LED, OUTPUT);//set pin as output Pin
  digitalWrite(pin_LED, LOW);//set pin as LOW
  pinMode(pin_buzzer, OUTPUT);//set pin as output Pin
  digitalWrite(pin_buzzer, LOW);//set pin as LOW

}

const int nmeas = 256; //NO. OF measurements
long int sumsum = 0; //running sum of 64 sums
long int skip = 0; //TOTAL  skipped sums
long int diff = 0;      //difference between sum and avgsum
long int flash_period = 0; //period (in ms)
long unsigned int prev_flash = 0; //time stamp of previous flash

void loop() {
  while (Serial.available() > 0)
  {
  val = Serial.read();
  Serial.println(val);
  }
```

```
if( val == 'F') // Forward MOVEMENT
  {
    digitalWrite(m1a, HIGH);
    digitalWrite(m1b, LOW);
    digitalWrite(m2a, HIGH);
    digitalWrite(m2b, LOW);
  }
else if(val == 'B') // Backward MOVEMENT
  {
    digitalWrite(m1a, LOW);
    digitalWrite(m1b, HIGH);
    digitalWrite(m2a, LOW);
    digitalWrite(m2b, HIGH);
  }

  else if(val == 'L') //Left MOVEMENT
  {
  digitalWrite(m1a, LOW);
  digitalWrite(m1b, LOW);
  digitalWrite(m2a, HIGH);
  digitalWrite(m2b, LOW);
  }
  else if(val == 'R') //Right MOVEMENT
  {
  digitalWrite(m1a, HIGH);
  digitalWrite(m1b, LOW);
  digitalWrite(m2a, LOW);
  digitalWrite(m2b, LOW);
  }

else if(val == 'S') //Stop MOVEMENT
  {
  digitalWrite(m1a, LOW);
  digitalWrite(m1b, LOW);
  digitalWrite(m2a, LOW);
  digitalWrite(m2b, LOW);
  }
else if(val == 'I') //Forward Right MOVEMENT
  {
  digitalWrite(m1a, HIGH);
  digitalWrite(m1b, LOW);
  digitalWrite(m2a, LOW);
  digitalWrite(m2b, LOW);
  }
else if(val == 'J') //Backward Right MOVEMENT
  {
  digitalWrite(m1a, LOW);
  digitalWrite(m1b, HIGH);
  digitalWrite(m2a, LOW);
  digitalWrite(m2b, LOW);
  }
 else if(val == 'G') //Forward Left MOVEMENT
  {
  digitalWrite(m1a, LOW);
  digitalWrite(m1b, LOW);
```

```
  digitalWrite(m2a, HIGH);
  digitalWrite(m2b, LOW);
  }
else if(val == 'H') //Backward Left MOVEMENT
  {
  digitalWrite(m1a,  LOW);
  digitalWrite(m1b,  LOW);
  digitalWrite(m2a,  LOW);
  digitalWrite(m2b, HIGH);
  }


int minval = 2000;
int maxval = 0;

//perform measurement
long unsigned int sum = 0;
for (int imeas = 0; imeas < nmeas + 2; imeas++) {
  //reset the capacitor
  pinMode(pin_cap, OUTPUT);
  digitalWrite(pin_cap, LOW);
  delayMicroseconds(20);
  pinMode(pin_cap, INPUT);
  //apply pulses
  for (int ipulse = 0; ipulse < npulse; ipulse++) {
    digitalWrite(pin_pulse, HIGH); //takes 3.5 microseconds
    delayMicroseconds(3);
    digitalWrite(pin_pulse, LOW); //takes 3.5 microseconds
    delayMicroseconds(3);
  }
  //read the charge on the capacitor
  int val = analogRead(pin_cap); //takes 13x8=104 microseconds
  minval = min(val, minval);
  maxval = max(val, maxval);
  sum += val;

  //determine if LEDs should be on or off
  long unsigned int timestamp = millis();
  byte ledstat = 0;
  if (timestamp < prev_flash +12) {
    if (diff > 0)ledstat = 1;
    if (diff < 0)ledstat = 2;
  }
  if (timestamp > prev_flash + flash_period) {
    if (diff > 0)ledstat = 1;
    if (diff < 0)ledstat = 2;
    prev_flash = timestamp;
  }
  if (flash_period > 1000)ledstat = 0;

  //switch the LEDs to this setting
  if (ledstat == 0) {
    digitalWrite(pin_LED, LOW);
    noTone(pin_buzzer);
  }
```

```
  if (ledstat == 1) {
    digitalWrite(pin_LED, LOW);
    noTone(pin_buzzer);
  }
  if (ledstat == 2) {
    digitalWrite(pin_LED, HIGH);
    tone(pin_buzzer,100);
  }
}

//subtract minimum and maximum value to remove spikes
sum -= minval; sum -= maxval;

//process
if (sumsum == 0) sumsum = sum << 6; //set sumsum to expected value
long int avgsum = (sumsum + 32) >> 6;
diff = sum - avgsum;
if (abs(diff)<avgsum >> 10) {   //adjust for small changes
  sumsum = sumsum + sum - avgsum;
  skip = 0;
} else {
  skip++;
}
if (skip > 64) {  // break off in case of prolonged skipping
  sumsum = sum << 6;
  skip = 0;
}

// one permille change = 2 ticks/s
if (diff == 0) flash_period = 1000000;
else flash_period = avgsum / (2 * abs(diff));
```

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